

# SAR Ship Detection - YOLO vs DETR

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# Outline

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# Introduction

- YOLO has become a mainstay for its speed with decent accuracy
  - Quick, easy implementation
  - Real-time testing
  - But, poor small-object detection, detail struggles
- A real-time transformer competitor, RT-DETR
  - More sensitive and accurate
- Ultralytics provides easy access - let's compare

# Dataset

SAR Ship Detection  
Dataset - SSDD [1]

1160 SAR images, inshore  
and offshore

Training and Testing sets  
already split

Annotated in COCO  
format

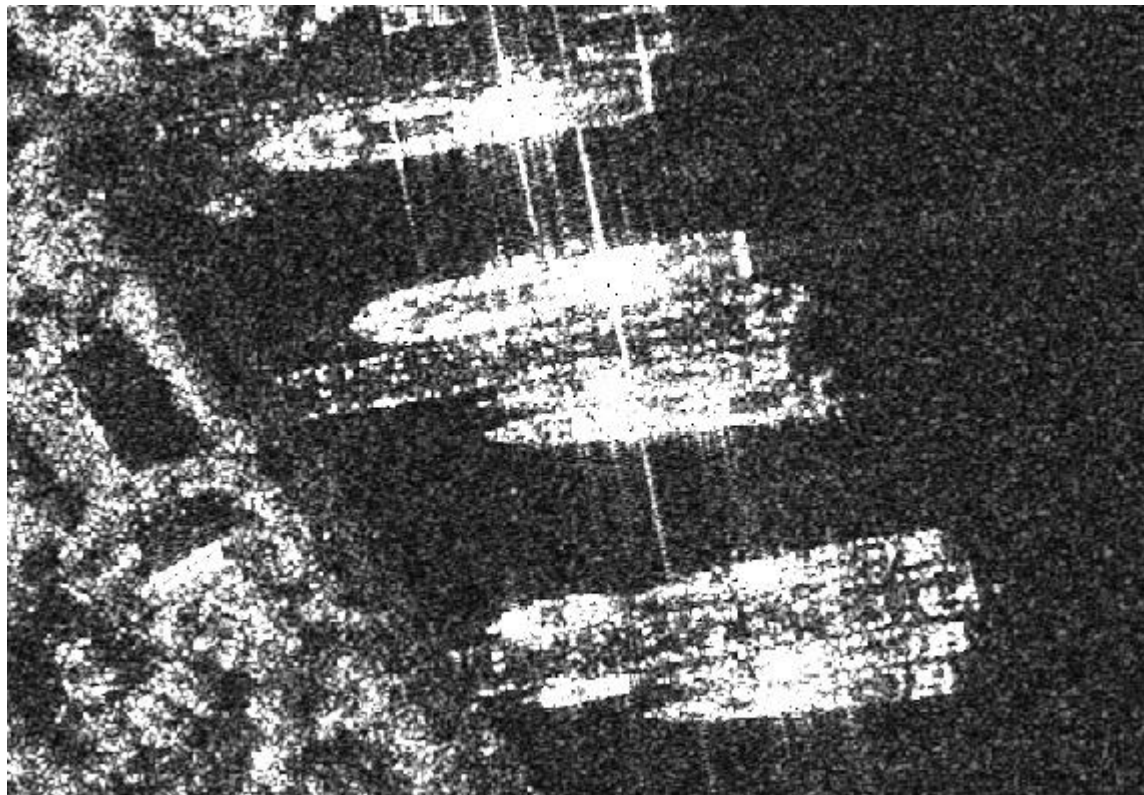


Figure 1. Image 000219.jpg from the SSDD [1]

# Models - YOLO26

CNN backbone

FPN, PAN neck to  
extract data across  
resolutions  
(feature fusion)

Detection heads  
generate anchors  
in each grid cell

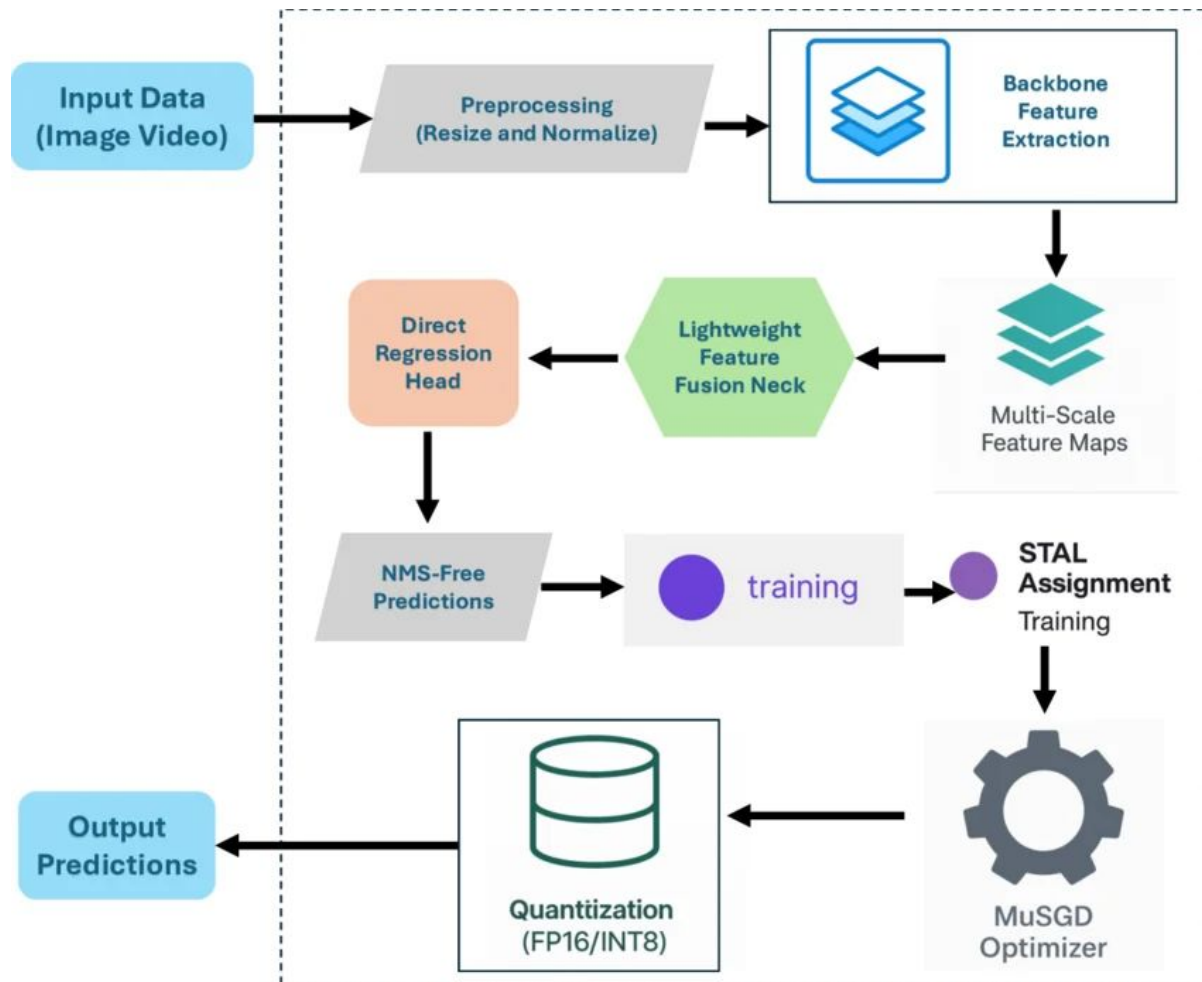


Figure 2. YOLO26 architecture as shown in the research paper [2]

# Models - RT-DETR

CNN backbone

Encode 3 layers -  
self-attend

Decode into  
queries,  
self-attend

Match objects to  
queries

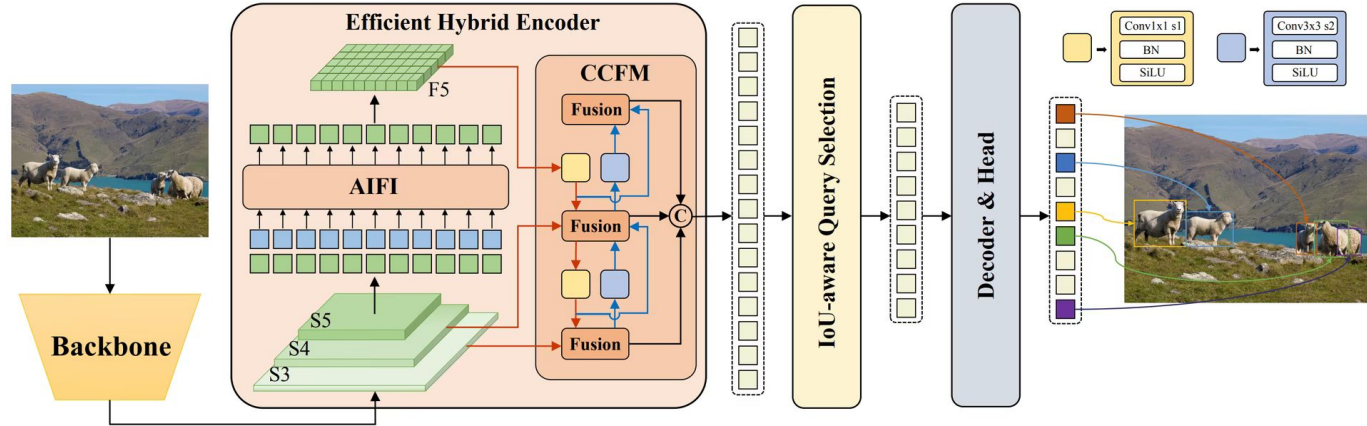


Figure 3 . RT-DETR architecture as shown in the research paper [3]

# Models - Activation and Loss Functions

## YOLO26

- Activation:
  - Objectness - sigmoid
  - Classification - sigmoid
- Loss:
  - Bounding Box Loss - IoU
  - Classification Loss - Cross Entropy
  - Objectness Loss - Cross Entropy

## RT-DETR

- Activation:
  - Bounding Box - sigmoid
  - Classification - softmax
- Loss:
  - Bipartite matching first! (Hungarian)
  - Bounding Box Loss - L1 and GloU
  - Classification Loss - Cross Entropy

# Tutorial Methods

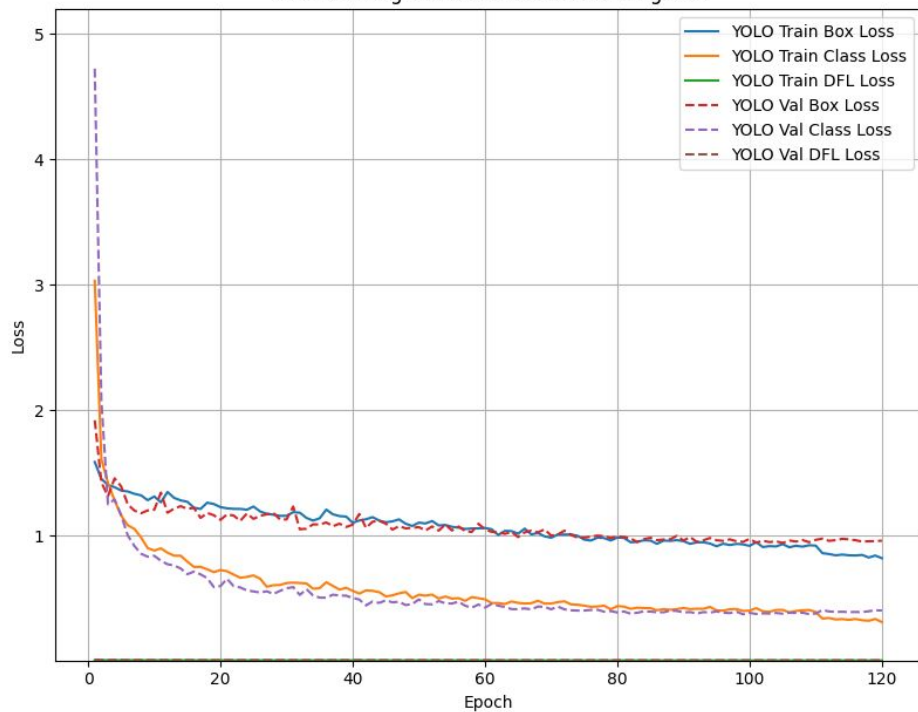
- Demonstrate training, optimization, testing on both models
  - Full training (120 epochs YOLO, 150 DETR)
  - Brief optimization (10 epochs, demonstrate loss regression)
  - Testing
- Compare results
  - Train/Val Loss Graphs
  - Confusion Matrices
  - Precision, Recall, F1



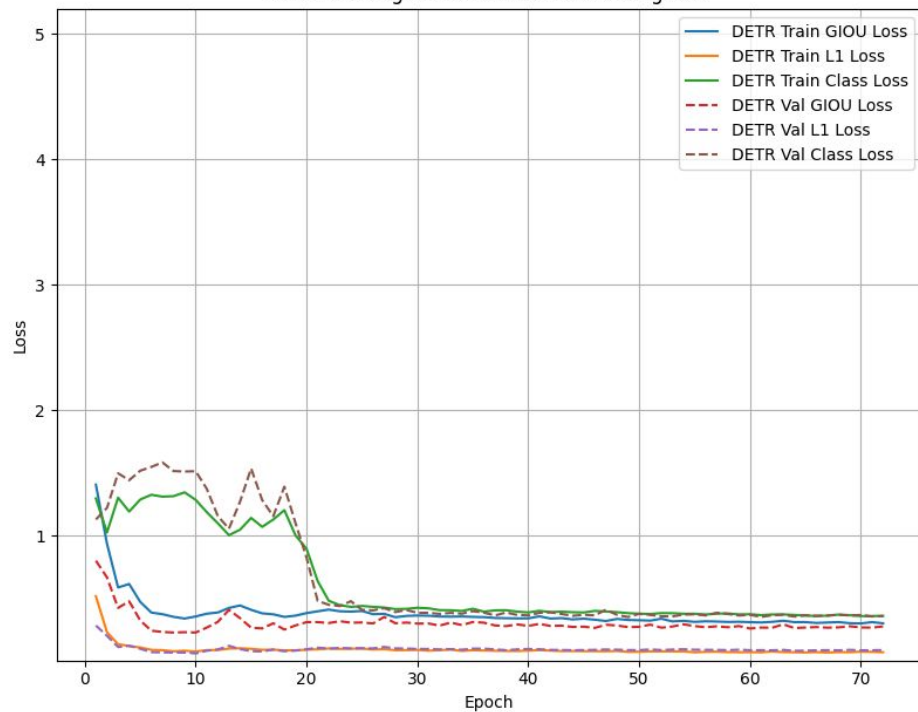
# Comparisons

Training Loss Convergence Comparison

YOLO Training & Validation Loss Convergence

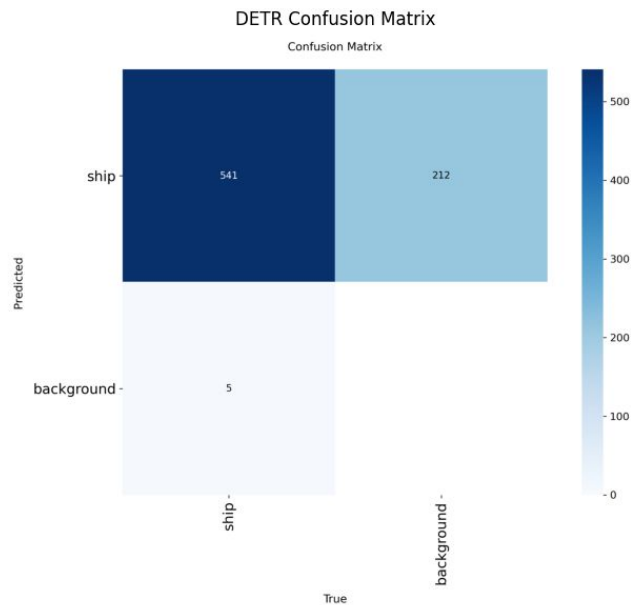
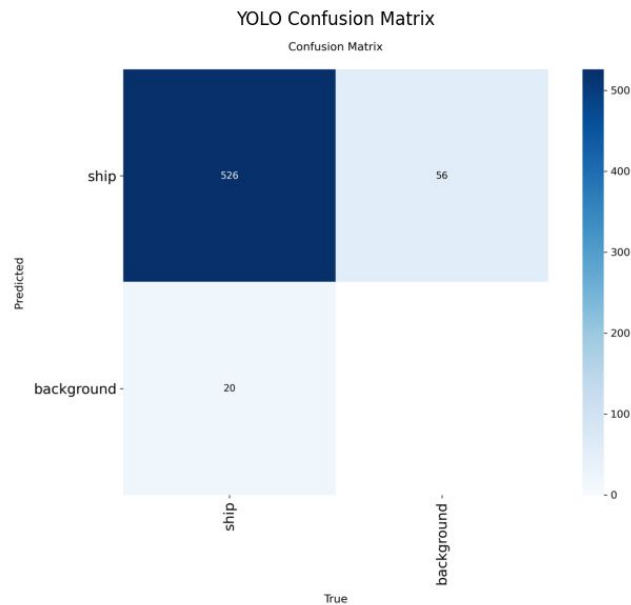


DETR Training & Validation Loss Convergence



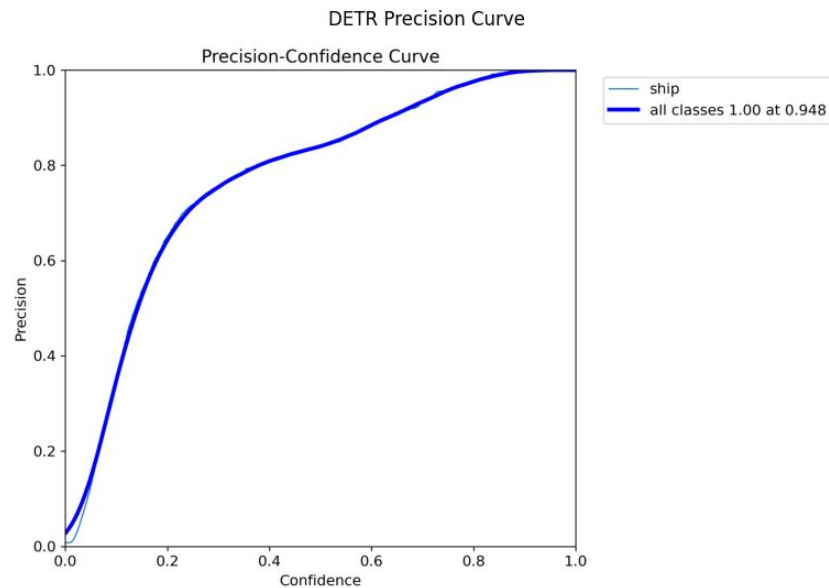
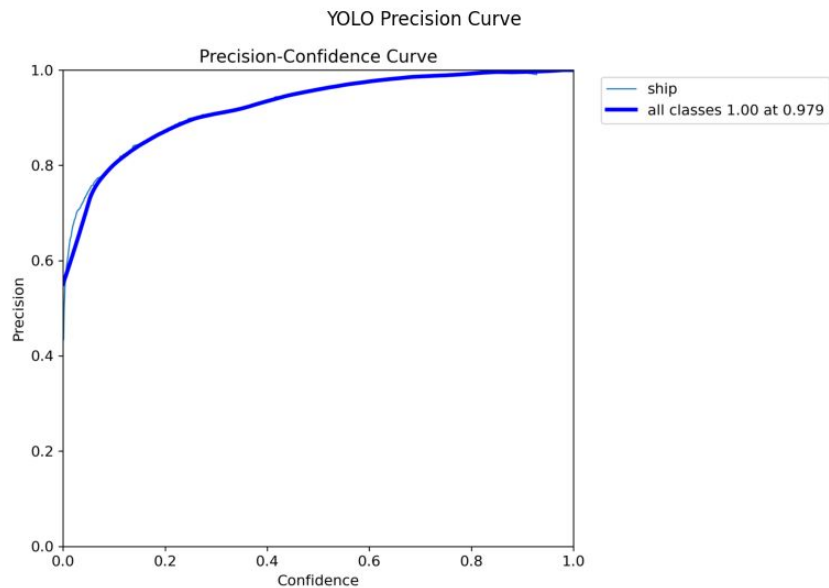
# Comparisons

## Confusion Matrix Comparison



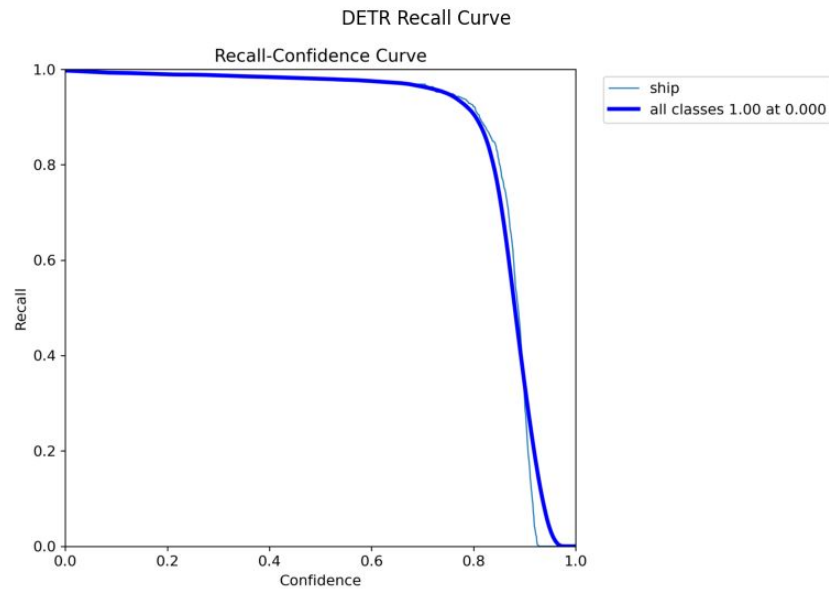
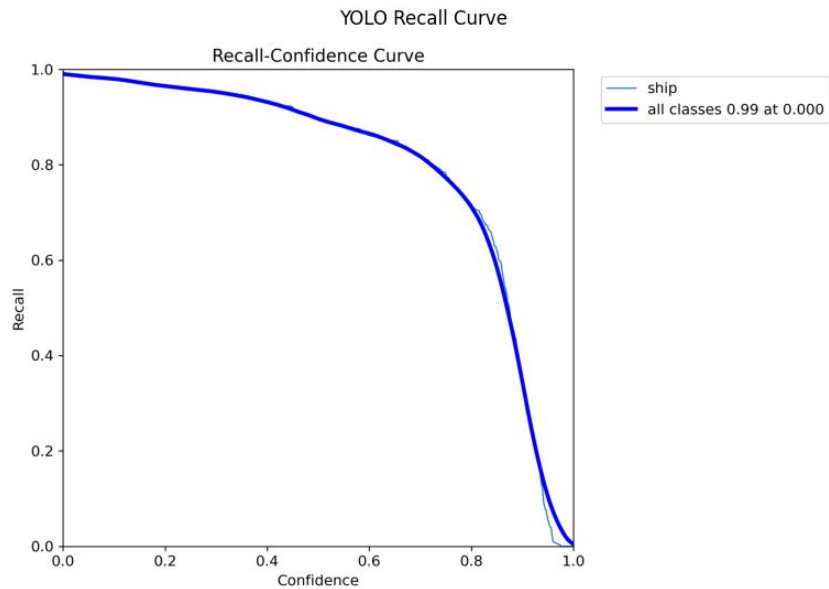
# Comparisons

## Precision Curve Comparison



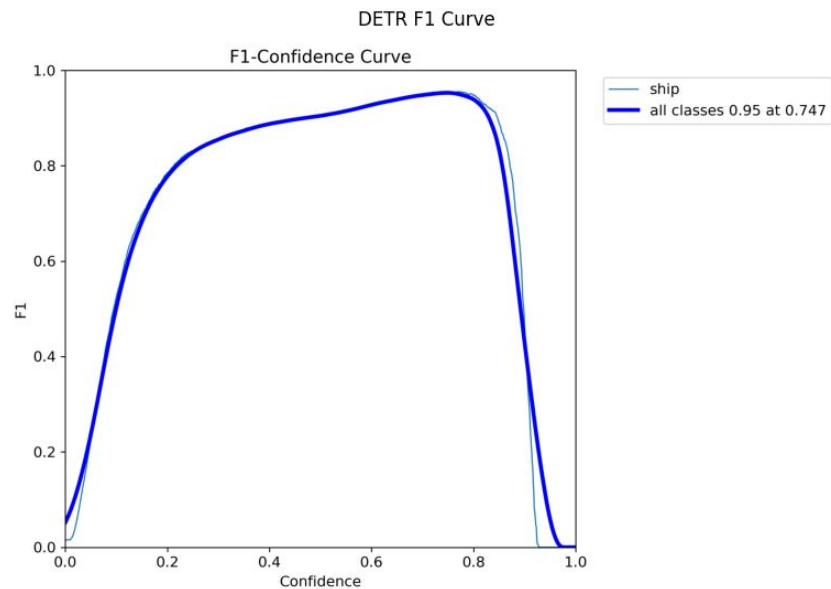
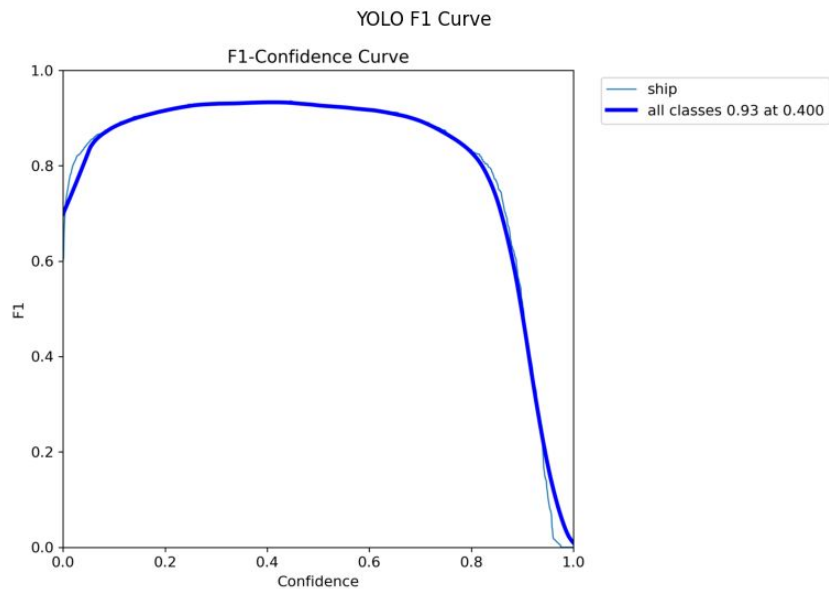
# Comparisons

## Recall Curve Comparison



# Comparisons

F1 Curve Comparison



# Conclusions

- An imperfect way to demonstrate this application, but useful
- Models are developing quickly both in usefulness and accessibility

# References

- [1] T. Zhang, X. Zhang, J. Li, X. Xu, B. Wang, X. Zhan, Y. Xu, X. Ke, T. Zeng, H. Su et al., “SAR ship detection dataset (SSDD): Official release and comprehensive data analysis,” Remote Sensing, vol. 13, no. 18, p. 3690, 2021, doi: 10.3390/rs13183690.
- [2] R. Sapkota and M. Karkee, “Ultralytics YOLO Evolution: An Overview of YOLO26, YOLO11, YOLOv8 and YOLOv5 Object Detectors for Computer Vision and Pattern Recognition,” arXiv:2510.09653, 2025. doi: 10.48550/arXiv.2510.09653.
- [3] W. Lv et al., “DETRs Beat YOLOs on Real-time Object Detection,” arXiv:2304.08069v3, 2024. doi: 10.48550/arXiv.2304.08069

**Questions?**